

- Web-based materials/chemistry simulation platform

Materials Square Class Curriculum

○



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[Lab.1] Surface Structures/ Surfaces Energies

Learning Objectives:

- Create surface structures according to various miller indices and check changes in surface structure through structural optimization calculations. Additionally, the stability of the surface is compared by calculating the surface energy.

Materials: Pt surfaces



MatSQ Credit: ~ 12 CPU hour (\$ 3.0)

[Lab.1] Surface Structures/ Surfaces Energies

Pt surfaces (low index)

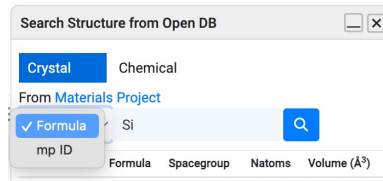
1. Build crystal structure
2. Calculate variable cell relaxation of bulk structure (primitive cell)
3. Build conventional cell
4. Calculate total energy of Pt conventional cell
5. Build Pt(111) surface structure and optimize the atomic structure
6. Build Pt(110) surface structure and optimize the atomic structure
7. Calculate atomic structure optimization
8. Calculate surface energies

MatSQ Credit: ~ 12 CPU hour (\$ 3.0)

[Lab.1] Surface Structures/ Surfaces Energies

1. Build crystal structure (DB, Load File, Preset)

① Using OpenDB

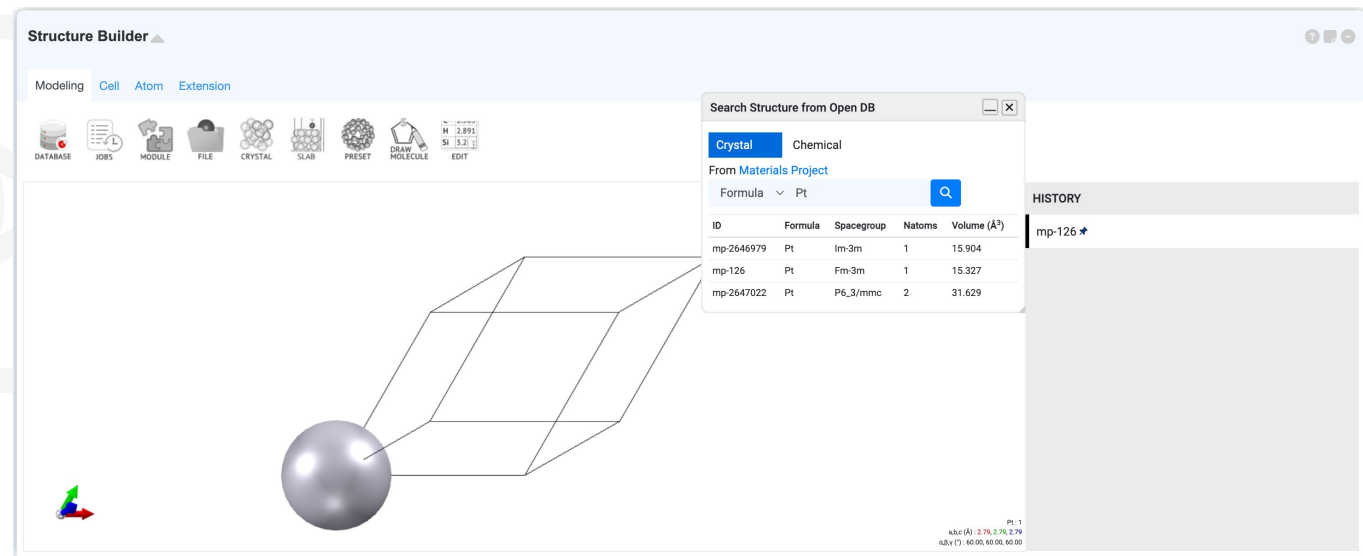
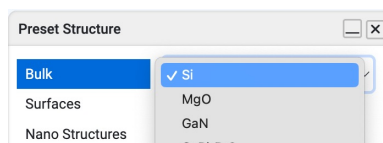


② Load files



.pdb .cif .xyz
...

③ Preset



[Lab.1] Surface Structures/ Surfaces Energies

2. Calculate variable cell relaxation of bulk structure (primitive cell)

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Bulk

Ionic+cell optimization

☐ Spin polarization

☐ Van der Waals

☐ Electric field

☒ Spin Orbit Coupling

☐ DFT+U

&CONTROL

Calculation type

vc-relax

Max SCF steps

200

Information amount

high

Force threshold (Ry/bohr)

0.00038

&SYSTEM

$E_{\text{cut}}(\text{wfc})(\text{Ry})$

40

$E_{\text{cut}}(\text{rho})(\text{Ry})$

320

Occupations

smearing

Gaussian broadening (Ry)

0.002

&ELECTRONS

Max iteration step

200

SCF must converge

True

Mixing mode

plain

Mixing beta

0.5

Convergence threshold (Ry)

0.000001

Starting wavefunction

random

KPOINTS

Sampling

Monkhorst-Pack

Grid

Shift

11111

000

&IONS

Ion dynamics

bfgs

Upscale

100

&CELL

Cell dynamics

bfgs

Cell factor

2

Pressure threshold (Kbar)

0.2

Cell relax type

all

[Lab.1] Surface Structures/ Surfaces Energies

3. Build conventional cell

Structure Builder

Modeling **Cell** Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Unit: Angstrom(Å)

Cell ▾ Lattice Parameters ▾

a	2.809534623	α (°)	60
b	2.809534623	β (°)	60
c	2.809534623	γ (°)	60

Atoms ▾ Cartesian ▾ ☐ Selected Atoms

ID	EL.	x	y	z
1	Pt	0	0	0

+ Add Atom

Reset Apply

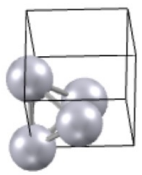
[Lab.1] Surface Structures/ Surfaces Energies

3. Build conventional cell

Structure Builder

Modeling Cell Atom Extension

STRAIN CLONE VACUUM CLEAVE MERGE PRIMITIVE CONVENTIONAL



HISTORY

- mp-126
- 147104-Final
- Manual Edit
- Conventional

Pt : 4
a,b,c (Å) : 3.97, 3.97, 3.97
 α,β,γ (°) : 90.00, 90.00, 90.00

[Lab.1] Surface Structures/ Surfaces Energies

4. Calculate total energy of Pt conventional cell

Quantum Espresso

PWscf + DOS ▼

Scripting Option : Template ▼ Atom info Update Structure Keyword information

Target property
Energy / Structure ▼

Precision
High (long time) ▼

Model type ⓘ
Bulk ▼

Structure optimization
No optimization ▼

Option
☐ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☐ DFT+U

&CONTROL
Calculation type scf ▼
Information amount high ▼
Force threshold (Ry/bohr) 0.00038

&SYSTEM
 $E_{\text{cut}}(\text{wfc})(\text{Ry})$ 40
 $E_{\text{cut}}(\text{rho})(\text{Ry})$ 320
Occupations smearing ▼
Gaussian broadening (Ry) 0.002

&ELECTRONS
Max iteration step 200
SCF must converge True ▼
Mixing mode plain ▼
Mixing beta 0.5
Convergence threshold (Ry) 0.000001
Starting wavefunction random ▼
KPOINTS
Sampling Monkhorst-Pack ▼
Grid 8 8 8
Shift 0 0 0

[Lab.1] Surface Structures/ Surfaces Energies

5-1. Build Pt(111) surface structure

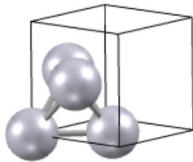
Structure Builder

Modeling Cell Atom Extension

STRAIN CLONE VACUUM CLEAVE MERGE PRIMITIVE CONVENTIONAL

Press ESC key or touch **THIS** to exit select mode

Select mode - Rectangular Normal



Cleave Surface

(h,k,l) = 1 1 1

(max c length Å) 30 ✓ Apply

no.	a	b	c	α	β	γ
1	5.619	5.619	5.619	60.000	60.000	60.000
2	5.619	5.619	9.733	73.221	90.000	60.000
3	5.619	5.619	13.764	90.000	90.000	60.000
4	5.619	5.619	18.636	90.000	98.671	60.000
5	5.619	5.619	23.168	83.035	90.000	60.000
6	5.619	5.619	28.095	84.261	95.739	60.000

a,b,c (Å) : 3.97, 3.97, 3.97
 α,β,γ (°) : 90.00, 90.00, 90.00

[Lab.1] Surface Structures/ Surfaces Energies

5-1. Build Pt(111) surface structure

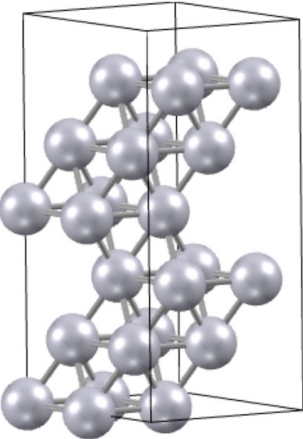
Structure Builder

Modeling Cell Atom Extension

STRAIN CLONE VACUUM CLEAVE MERGE PRIMITIVE CONVENTIONAL

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular Normal



HISTORY

New Structure

Cleave (1 1 1)

Pt: 24
a,b,c (Å): 5.62, 5.62, 13.76
α,β,γ (°): 90.00, 90.00, 60.00

[Lab.1] Surface Structures/ Surfaces Energies

5-1. Build Pt(111) surface structure

Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular Normal

Manual Edit

b 5.6190615219 β (°) 90

c 25 γ (°) 59.999999999

Atoms Cartesian ☐ Selected Atoms

ID	EL.	x	y	z
19	Pt	4.2142	2.4331	0
20	Pt	1.4047	0.8110	2.2939
21	Pt	7.0238	4.0552	4.5879
22	Pt	4.2142	2.4331	6.8819
23	Pt	1.4047	0.8110	9.1758

+ Add Atom

Reset Apply

[Lab.1] Surface Structures/ Surfaces Energies

5-2. Optimize atomic structure of Pt(111) surface structures

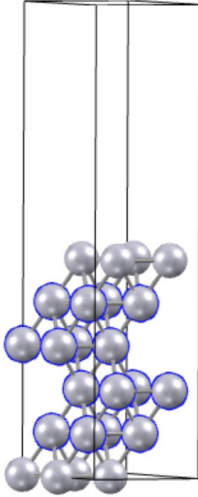
Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular
Normal
Pt=16



HISTORY

- New Structure
- Cleave (1 1 1)
- Manual Edit

Pt: 24
a,b,c (Å): 5.62, 5.62, 25.00
α,β,γ (°): 90.00, 90.00, 60.00

[Lab.1] Surface Structures/ Surfaces Energies

5-2. Optimize atomic structure of Pt(111) surface structures

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info

Target property
Energy / Structure

&CONTROL
Calculation type
scf

input.pw.x Potential

	Checkbox	Input	Selected
Fix			
No.	Ele.	☐	☐ x ☐ y ☐ z
1	Pt	☐	☐ 0.000 ☐ 0.000 ☐ 0.000
2	Pt	☒	☒ 5.619 ☒ 3.244 ☒ 2.294
3	Pt	☒	☒ 2.810 ☒ 1.622 ☒ 4.588
4	Pt	☒	☒ 0.000 ☒ 0.000 ☒ 6.882
5	Pt	☒	☒ 5.619 ☒ 3.244 ☒ 9.176
6	Pt	☐	☐ 2.810 ☐ 1.622 ☐ 11.470
7	Pt	☐	☐ 2.810 ☐ 0.000 ☐ 0.000
8	Pt	☒	☒ 2.810 ☒ 3.244 ☒ 2.294

Keyword information

Structure optimization
No optimization

30
smearing
0.002

&ELECTRONS
Max iteration step
SCF must converge
Mixing beta
KPOINTS
Sampling
Grid
Shift

100
True
0.5
Monkhorst-Pack
3 3 1
0 0 0

Job Submit

[Lab.1] Surface Structures/ Surfaces Energies

5-2. Optimize atomic structure of Pt(111) surface structures

Quantum Espresso ▲

PWscf

+

DOS

▼

Scripting Option :

Template ▼

[Atom info](#) ☒ Update Structure [Keyword information](#)

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure ▼

High (long time) ▼

Isolate System ▼

Ionic optimization ▼

☐ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☐ DFT+U

&CONTROL

Calculation type

Max SCF steps

Information amount

Force threshold (Ry/bohr)

&IONS

Ion dynamics

Upscale

relax ▼

200

high ▼

0.00038

bfgs ▼

100

&SYSTEM

$E_{\text{cut}}(\text{wfc})(\text{Ry})$

$E_{\text{cut}}(\text{rho})(\text{Ry})$

Occupations

Gaussian broadening (Ry)

40

320

smearing ▼

0.002

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

KPOINTS

Sampling

Grid

Shift

200

True ▼

plain ▼

0.5

0.000001

random ▼

Monkhorst-Pack ▼

5

5

1

0

0

0

15

[Lab.1] Surface Structures/ Surfaces Energies

6-1. Build Pt(110) surface structure

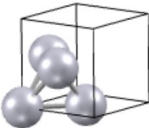
Structure Builder

Modeling Cell Atom Extension

STRAIN CLONE VACUUM CLEAVE MERGE PRIMITIVE CONVENTIONAL

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular Normal



Cleave Surface

(h,k,l) = 1 1 0

(max c length Å) 30 ✓ Apply

no.	a	b	c	α	β	γ
1	5.619	3.973	3.973	90.000	45.000	90.000
2	5.619	3.973	5.619	90.000	90.000	90.000
3	5.619	3.973	8.885	90.000	71.565	90.000
4	5.619	3.973	11.238	90.000	90.000	90.000
5	5.619	3.973	14.326	90.000	78.690	90.000
6	5.619	3.973	16.857	90.000	90.000	90.000
7	5.619	3.973	19.866	90.000	81.870	90.000

Pt : 4
a,b,c (Å) : 3.97, 3.97, 3.97
 α,β,γ (°) : 90.00, 90.00, 90.00

[Lab.1] Surface Structures/ Surfaces Energies

6-1. Build Pt(110) surface structure

Structure Builder

Modeling Cell Atom Extension

STRAIN CLONE VACUUM CLEAVE MERGE PRIMITIVE CONVENTIONAL

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular Normal

Clone (# of Cell)

☒ Clone with cell ☐ Paste selected

a b c

1 1 3

☐ Clone selected atom(s)

✓ Apply

Pt : 8
a,b,c (Å) : 5.62, 3.97, 5.62
α,β,γ (°) : 90.00, 90.00, 90.00

[Lab.1] Surface Structures/ Surfaces Energies

6-1. Build Pt(110) surface structure

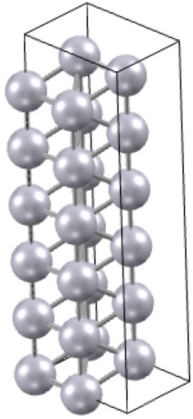
Structure Builder

Modeling Cell Atom Extension

STRAIN CLONE VACUUM CLEAVE MERGE PRIMITIVE CONVENTIONAL

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular Normal



Clone (# of Cell)

☒ Clone with cell ☐ Paste selected

a b c

1 1 3

☐ Clone selected atom(s)

✓ Apply

Pt : 24
a,b,c (Å) : 5.62, 3.97, 16.86
α,β,γ (°) : 90.00, 90.00, 90.00

[Lab.1] Surface Structures/ Surfaces Energies

6-1. Build Pt(110) surface structure

Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular Normal

Manual Edit

b 3.97327650605 β (°) 90

c 30 γ (°) 90

Atoms Cartesian ☐ Selected Atoms

ID	EL.	x	y	z
1	Pt	0	0	0
2	Pt	2.80955	0	2.80955
3	Pt	2.80955	0	0
4	Pt	0	0	2.80955
5	Pt	4.21425	1.98665	1.40475

+ Add Atom

Reset Apply

[Lab.1] Surface Structures/ Surfaces Energies

6-2. Optimize atomic structure of Pt(110) surface structures

Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular
Normal
Pt=16

HISTORY

- New Structure
- Cleave (1 1 0)
- Clone
- Manual Edit

Pt: 24
a,b,c (Å): 5.62, 3.97, 30.00
α,β,γ (°): 90.00, 90.00, 90.00

The screenshot displays the 'Structure Builder' software interface. The main window shows a 3D model of a Pt(110) surface, which is a rectangular slab of platinum atoms. The atoms are represented as blue spheres. The model is set in 'Normal' mode with a rectangular cell. The 'HISTORY' panel on the right shows a list of actions: 'New Structure', 'Cleave (1 1 0)', 'Clone', and 'Manual Edit'. The bottom status bar provides details about the model: 'Pt: 24', 'a,b,c (Å): 5.62, 3.97, 30.00', and 'α,β,γ (°): 90.00, 90.00, 90.00'. The top navigation bar includes tabs for 'Modeling', 'Cell', 'Atom', and 'Extension'. The left sidebar contains icons for 'DATABASE', 'JOBS', 'MODULE', 'FILE', 'CRYSTAL', 'SLAB', 'PRESET', 'DRAW MOLECULE', and 'EDIT'.

[Lab.1] Surface Structures/ Surfaces Energies

6-2. Optimize atomic structure of Pt(110) surface structures

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info

Target property
Energy / Structure

&CONTROL

Calculation type
scf

input.pw.x

Potential

	Checkbox	Input	Selected
		Fix	
No.	Ele.	☐ x ☐ y ☐ z	
1	Pt	☐ 0.000 ☐ 0.000 ☐ 0.000	
2	Pt	☒ 2.810 ☒ 0.000 ☒ 2.810	
3	Pt	☐ 2.810 ☐ 0.000 ☐ 0.000	
4	Pt	☒ 0.000 ☒ 0.000 ☒ 2.810	
5	Pt	☐ 4.214 ☐ 1.987 ☐ 1.405	
6	Pt	☒ 1.405 ☒ 1.987 ☒ 4.214	
7	Pt	☐ 1.405 ☐ 1.987 ☐ 1.405	
8	Pt	☒ 4.214 ☒ 1.987 ☒ 4.214	

Job Submit

Keyword information

Structure optimization
No optimization

&ELECTRONS

Max iteration step
100

SCF must converge
True

Mixing beta
0.5

KPOINTS

Sampling
Monkhorst-Pack

Grid
3 4 1

Shift
0 0 0

[Lab.1] Surface Structures/ Surfaces Energies

6-2. Optimize atomic structure of Pt(110) surface structures

Quantum Espresso

PWscf + DOS ▼

Scripting Option : Template ▼ Atom info Update Structure Keyword information

Target property
Energy / Structure ▼

Precision
High (long time) ▼

Model type ⓘ
Isolate System ▼

Structure optimization
Ionic optimization ▼

Option
☐ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☐ DFT+U

&CONTROL
Calculation type relax ▼
Max SCF steps 200
Information amount high ▼
Force threshold (Ry/bohr) 0.00038

&SYSTEM
 $E_{\text{cut}}(\text{wfc})(\text{Ry})$ 40
 $E_{\text{cut}}(\text{rho})(\text{Ry})$ 320
Occupations smearing ▼
Gaussian broadening (Ry) 0.002

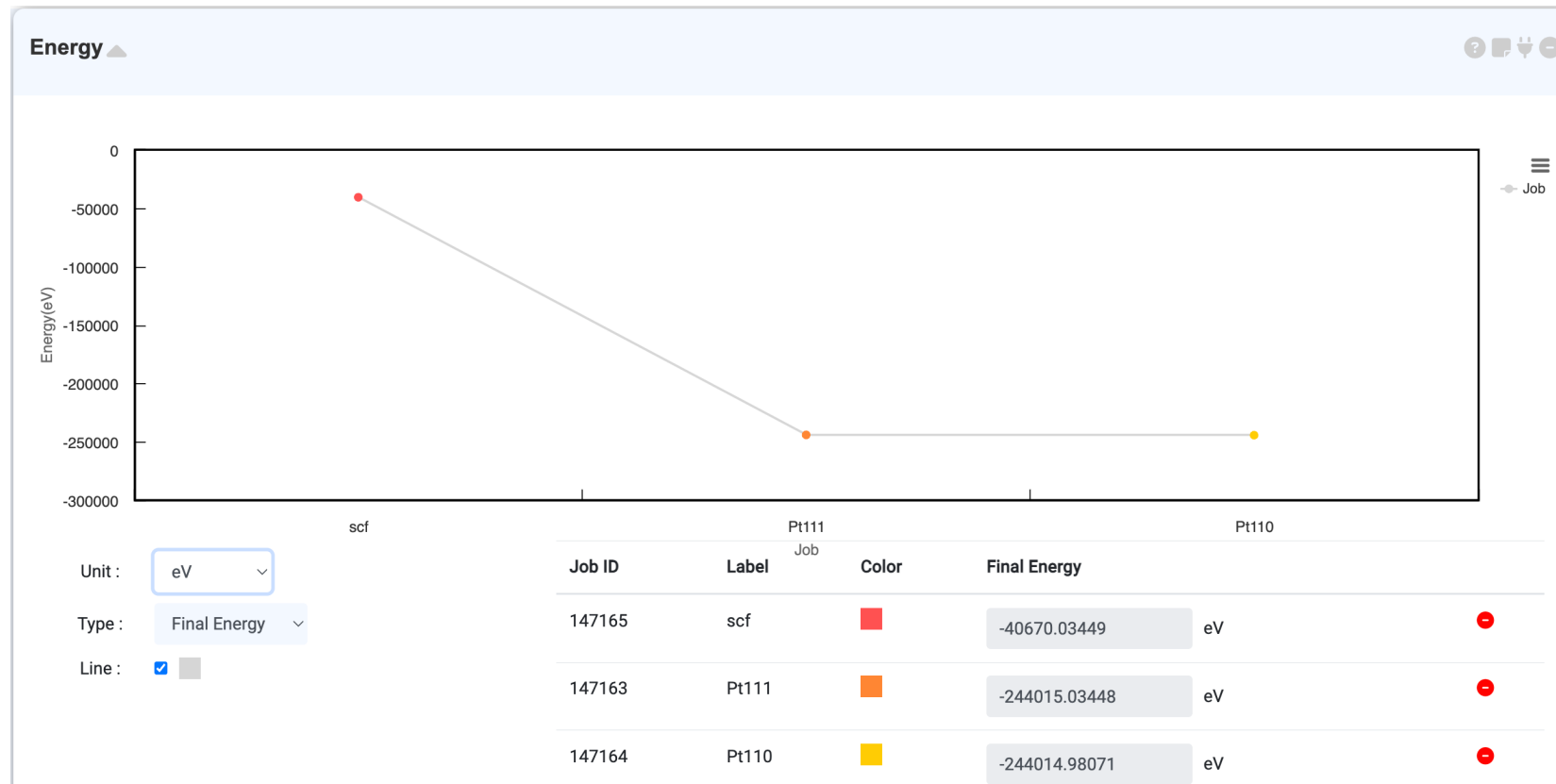
&ELECTRONS
Max iteration step 200
SCF must converge True ▼
Mixing mode plain ▼
Mixing beta 0.5
Convergence threshold (Ry) 0.000001
Starting wavefunction random ▼
KPOINTS
Sampling Monkhorst-Pack ▼
Grid 5 8 1
Shift 0 0 0

&IONS
Ion dynamics bfgs ▼
Upscale 100

[Lab.1] Surface Structures/ Surfaces Energies

7. Calculate surface energy: Pt surface

$$E_{surface} = \frac{1}{2} \times \frac{(E_{slab} - N \times E_{unitcell})}{Surface Area}$$



[Lab.1] Surface Structures/ Surfaces Energies

7. Calculate surface energy: Pt surface

$$E_{surface} = \frac{1}{2} \times \frac{(E_{slab} - N \times E_{unitcell})}{Surface\ Area}$$

Memo ▲

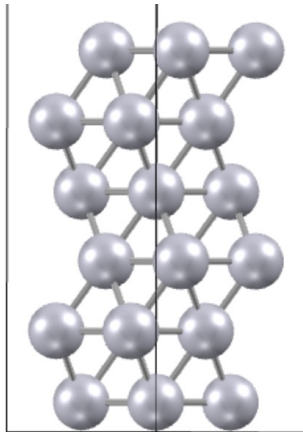
Note Sheet

	A ▼	B ▼	C ▼	D ▼	E ▼	F ▼	G ▼	H ▼	I ▼	J ▼	K ▼
1		# of atoms	Total Energy (eV)	Surface Area (Å²)	Surface Energy (eV/Å²)						
2	Pt-bulk	4	-40670.034								
3	Pt(111)	24	-244015.00	27.343159	0.0945841						
4	Pt(110)	24	-244014.90	22.324286	$=(C4-6*C2)/(2*D4)$						
5											
6											
7											
8											
9											
10											

[Lab.1] Surface Structures/ Surfaces Energies

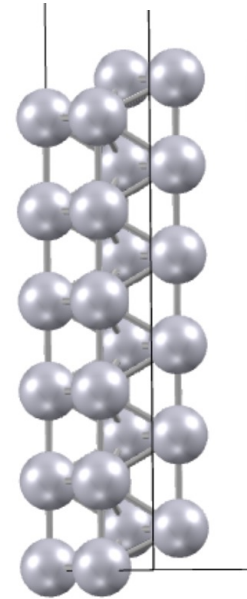
7. Calculate surface energy: Pt surface

Pt(111)



0.094 eV/Å^2

Pt(110)



0.117 eV/Å^2